

# Introduction to Limits\*

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## 1 Definition

Let  $y = f(x)$  be a function, and let  $a$  and  $L$  be numbers. The limit of  $f(x)$  as  $x$  approaches  $a$  is  $L$  if  $f(x)$  can be made close enough to  $L$  by taking  $x$  close enough to  $a$  (but not equal to  $a$ ). In other words, limits describe how a function behaves near a point, instead of at that point. That is,

$$\lim_{x \rightarrow a} f(x) = L$$

Note: What happens at  $x = a$  is irrelevant.

For example, if we want to find the limit of  $f(x) = 3x + 1$  as  $x$  approaches 0, we are interested in finding out what value  $f(x)$  takes on as  $x$  gets closer and closer to 0. We aren't interested in what actually happens at  $x = 0$ .

" $x$  close enough to  $a$ " must be on both sides.  $y$  has to approach  $L$  as  $x$  approaches  $a$  with  $x < a$  and with  $x > a$ . That is,

$$\lim_{x \rightarrow a^-} f(x) = L = \lim_{x \rightarrow a^+} f(x)$$

The first limit is "the limit of  $f(x)$  as  $x$  approaches  $a$  from the left" and the second limit is "the limit of  $f(x)$  as  $x$  approaches  $a$  from the right". They both must equal each other for the limit to exist.

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\*Definitions and Examples from Dr. Donald White's Lecture

## 2 Limit Laws

1. If  $c$  is a constant, and  $a$  is a number, then:  $\lim_{x \rightarrow a} c = c$
2.  $\lim_{x \rightarrow a} x = a$
3.  $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$
4.  $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$
5. If  $c$  is a constant, and  $a$  is a number, then:  $\lim_{x \rightarrow a} cf(x) = c \lim_{x \rightarrow a} f(x)$
6. If  $n$  is a positive integer, then:  $\lim_{x \rightarrow a} x^n = a^n$
7.  $\lim_{x \rightarrow a} (f(x))^n = (\lim_{x \rightarrow a} f(x))^n$
8.  $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$  If  $\lim_{x \rightarrow a} g(x) \neq 0$

## 3 Examples

1.  $\lim_{x \rightarrow 1} (3x + 1)$

In this case, we can just plug 1 into our function to get our limit.

$$\lim_{x \rightarrow 1} (3x + 1) = 3(1) + 1 = 4$$

2.  $\lim_{x \rightarrow 1} \frac{2x^7 + 5x^2}{(4x^3 + 3)^5}$

$$\begin{aligned}
 & \lim_{x \rightarrow 1} \frac{2x^7 + 5x^2}{(4x^3 + 3)^5} \\
 &= \frac{\lim_{x \rightarrow 1} 2x^7 + \lim_{x \rightarrow 1} 5x^2}{\lim_{x \rightarrow 1} (4x^3 + 3)^5} \text{ (by limit law 8)} \\
 &= \frac{\lim_{x \rightarrow 1} 2x^7 + \lim_{x \rightarrow 1} 5x^2}{(\lim_{x \rightarrow 1} 4x^3 + \lim_{x \rightarrow 1} 3)^5} \text{ (by limit law 3 and 7)} \\
 &= \frac{\lim_{x \rightarrow 1} 2x^7 + \lim_{x \rightarrow 1} 5x^2}{(\lim_{x \rightarrow 1} 4x^3 + \lim_{x \rightarrow 1} 3)^5} \text{ (by limit law 3)} \\
 &= \frac{2 \lim_{x \rightarrow 1} x^7 + 5 \lim_{x \rightarrow 1} x^2}{(4 \lim_{x \rightarrow 1} x^3 + \lim_{x \rightarrow 1} 3)^5} \text{ (by limit law 5)} \\
 &= \frac{2(1)^7 + 5(1)^2}{(4(1)^3 + 3)^5} \text{ (by limit law 1)} \\
 &= \frac{1}{2401}
 \end{aligned}$$

$$3. \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

Here we want to factor out anything we can, so we will factor the numerator:

$$\begin{aligned} & \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} \\ &= \lim_{x \rightarrow 2} \frac{(x + 2)(x - 2)}{x - 2} \end{aligned}$$

Since  $x \neq 2$ , we can say that the  $\frac{(x - 2)}{x - 2}$  is equal to 1, and drop them from our limit. So,

$$\begin{aligned} &= \lim_{x \rightarrow 2} (x + 2) \\ &= (2 + 2) \\ &= 4 \end{aligned}$$